1(A). Show that

$$Z\left[\sum_{h=0}^{k} x(h)\right] = \frac{1}{1-z^{-1}}X(z),$$
$$Z\left[\sum_{h=0}^{k-1} x(k)\right] = \frac{z^{-1}}{1-z^{-1}}X(z).$$

[4 marks]

Solution.

– Let $y(k)$ be the accumulation of the sequence $x(h)$ up to time k:

$$y(k) = \sum_{h=0}^{k} x(h)$$

This can be written recursively as:

$$y(k) - y(k-1) = x(k)$$

Taking the Z-transform of both sides and using the time-shifting property $\mathcal{Z}[y(h-1)] = z^{-1}Y(z)$:

$$\begin{aligned} Y(z) - z^{-1}Y(z) &= X(z) \\ Y(z)(1-z^{-1}) &= X(z) \\ Y(z) &= \frac{1}{1-z^{-1}}X(z) \end{aligned}$$

Thus, we have shown:

$$\mathcal{Z}\left[\sum_{h=0}^{k} x(h)\right] = \frac{1}{1-z^{-1}}X(z)$$

– Let $w(k) = \sum_{h=0}^{k-1} x(h)$. Note that $w(k) = y(k-1)$, where $y(k)$ is the sum defined above. Using the time-shifting property on the result above, we have

$$\begin{aligned} W(z) &= z^{-1}Y(z) \\ W(z) &= z^{-1}\left(\frac{1}{1-z^{-1}}X(z)\right) \\ W(z) &= \frac{z^{-1}}{1-z^{-1}}X(z) \end{aligned}$$

Thus:

$$\mathcal{Z}\left[\sum_{h=0}^{k-1} x(h)\right] = \frac{z^{-1}}{1-z^{-1}}X(z)$$

1(B). Compute the inverse Z-transform of

$$X(z) = \frac{z(z+2)}{(z-1)^2}.$$

[4 marks]

Solution: We use the Partial Fraction Expansion method on $\frac{X(z)}{z}$.

$$\frac{X(z)}{z} = \frac{z+2}{(z-1)^2} = \frac{A}{z-1} + \frac{B}{(z-1)^2}$$

To find the coefficients:

$$z + 2 = A(z-1) + B$$
$$\text{Let } z = 1 \implies B = 3$$
$$\text{Coeff of } z : 1 = A \implies A = 1$$

Rewriting $X(z)$:

$$X(z) = \frac{z}{z-1} + 3\frac{z}{(z-1)^2}$$

Using standard transform pairs $\mathcal{Z}^{-1}\{\frac{z}{z-1}\} = u[k]$ and $\mathcal{Z}^{-1}\{\frac{z}{(z-1)^2}\} = k$, we have

$$x[k] = (1 + 3k)u[k]$$

1(B). Consider the rectangular region in the s-plane defined by

$$-\sigma_2 \leq Re(s) \leq -\sigma_1 \text{ and } -\omega_2 \leq Im(s) \leq \omega_1,$$

with $\sigma_2 > \sigma_1 > 0$, $\omega_s/2 > \omega_2 > 0$, and $\omega_s/2 > \omega_1 > 0$ with $\omega_2 \neq \omega_1$. Determine the image of this region in the z-plane resulting from the mapping $z = e^{sT}$. Here $\omega_s = \frac{2\pi}{T}$. [4 marks]

Solution:

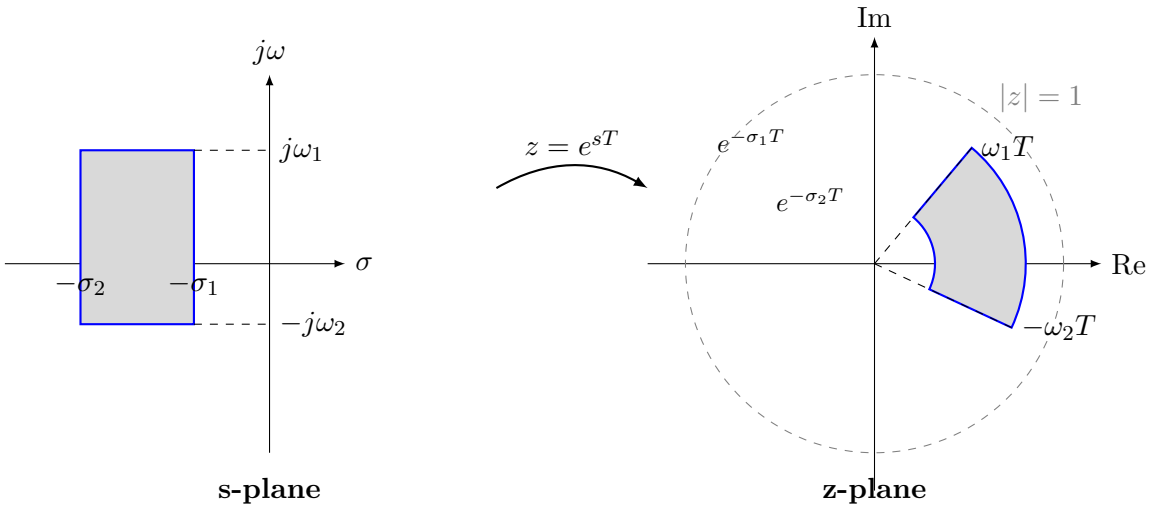


Figure 1: Mapping of the rectangular region from the s-plane to the z-plane.

1(D). Consider a position control system where the plant is a DC motor modeled by the transfer function:

$$P(s) = \frac{1}{s(s+1)}$$

(i) Suppose that the continuous-time system is controlled a proportional controller K. Determine for which values of K the closed-loop continuous-time system is asymptotically stable. [2 marks]

Now suppose that the system is controlled by a digital proportional controller K through a Zero-Order Hold (ZOH) with a sampling time $T_s = 1$ second.

(ii) Compute the equivalent discrete-time model $HP(z)$ for the plant with the hold and sampler. [3 marks]

(iii) Determine the range of K for which the closed-loop system is stable. [4 marks]

(iv) Sketch the root locus of the closed-loop system for $K > 0$. Your sketch must clearly indicate:

– The open-loop poles and zeros.
– The segments of the real axis that belong to the root locus.
– The breakaway point (calculate its approximate location).

[5 marks]

Solution.

(i) The system has a unity feedback loop with a proportional controller K. The closed-loop transfer function $F(s)$ is given by:

$$F(s) = \frac{KP(s)}{1 + KP(s)} \tag{1}$$

Substituting $P(s) = \frac{1}{s(s+1)}$:

$$\begin{aligned} F(s) &= \frac{K \cdot \frac{1}{s(s+1)}}{1 + K \cdot \frac{1}{s(s+1)}} \\ &= \frac{K}{s(s+1) + K} \\ &= \frac{K}{s^2 + s + K} \end{aligned}$$

The stability of the system depends on the roots of the characteristic equation (the denominator of $F(s)$):

$$\Delta(s) = s^2 + s + K = 0 \tag{2}$$

For a second-order polynomial of the form $as^2 + bs + c = 0$, the necessary and sufficient condition for asymptotic stability (all roots in the left-half plane) is that all coefficients a, b, c must be strictly positive. Therefore, for stability, we require:

$$K > 0 \tag{3}$$

(ii) Using the ZOH formula:

$$HP(z) = (1 - z^{-1})\mathcal{Z}\left\{\frac{1}{s^2(s+1)}\right\}$$

Perform partial fraction expansion on $\frac{1}{s^2(s+1)}$:

$$\frac{1}{s^2(s+1)} = \frac{1}{s^2} - \frac{1}{s} + \frac{1}{s+1}$$

Taking the Z-transform (using $T_s = 1$):

$$\mathcal{Z}\{\dots\} = \frac{1 \cdot z}{(z-1)^2} - \frac{z}{z-1} + \frac{z}{z-e^{-1}}$$

Multiplying by $(1 - z^{-1}) = \frac{z-1}{z}$:

$$HP(z) = \frac{(z - e^{-1}) - (z-1)(z-e^{-1}) + (z-1)^2}{(z-1)(z-e^{-1})}.$$

Simplifying the numerator with $e^{-1} \approx 0.368$:

$$\text{Num} = (z - 0.368) - (z^2 - 1.368z + 0.368) + (z^2 - 2z + 1)$$

$$\text{Num} = z - 0.368 - z^2 + 1.368z - 0.368 + z^2 - 2z + 1$$

$$\text{Num} = (1 + 1.368 - 2)z + (1 - 0.368 - 0.368) = 0.368z + 0.264$$

Thus, we have

$$HP(z) = \frac{0.368(z + 0.717)}{(z-1)(z-0.368)}$$

That is,

- Gain Factor $K_{dc} \approx 0.368$
- Zero $z_0 \approx 0.717$ (location $z = -0.717$)
- Pole $p_1 \approx 0.368$

(iii) The characteristic equation is

$$1 + K\frac{0.368(z+0.717)}{(z-1)(z-0.368)} = 0$$

$$(z-1)(z-0.368) + 0.368K(z+0.717) = 0$$

$$(z^2 - 1.368z + 0.368) + (0.368Kz + 0.263856K) = 0$$

The system is stable if

$$\begin{cases} 1 - (0.368K - 1.368) + 0.263856K + 0.368 > 0 \\ 0.263856K + 0.368 < 1 \\ 1 + (0.368K - 1.368) + 0.263856K + 0.368 > 0 \end{cases}$$

which gives

$$0 < K < 2.39.$$

(iv) **Step 1. Characteristic Equation** The closed-loop characteristic equation is given by $1+KG(z) = 0$:

$$(z-1)(z-0.368) + K \cdot 0.368(z+0.717) = 0 \tag{4}$$

Step 2. Open-Loop Poles and Zeros We identify the singularities of $G(z)$:

- **Poles ($n = 2$):** $p_1 = 1, \quad p_2 = 0.368$
- **Zeros ($m = 1$):** $z_1 = -0.717$

Since $n > m$, there is $n - m = 1$ branch that extends to infinity.

Step 3. Locus on the Real Axis The root locus exists on the real axis to the left of an odd number of poles and zeros.

- **Segment 1:** $0.368 < z < 1$ (Between the two poles).
- **Segment 2:** $z < -0.717$ (To the left of the zero).

Step 4. Breakaway and Break-in Points We solve for the points where the locus leaves or enters the real axis by setting $\frac{dK}{dz} = 0$. Rearranging the characteristic equation for K:

$$K = -\frac{(z-1)(z-0.368)}{0.368(z+0.717)} \tag{5}$$

Differentiating and setting to zero:

$$\frac{d}{dz}\left(\frac{z^2 - 1.368z + 0.368}{z + 0.717}\right) = 0 \tag{6}$$

Applying the quotient rule yields the quadratic equation:

$$z^2 + 1.434z - 1.349 = 0 \tag{7}$$

Solving for z:

$$z_{1,2} = \frac{-1.434 \pm \sqrt{(1.434)^2 - 4(1)(-1.349)}}{2} \tag{8}$$

This provides two solution points:

$$\sigma_b \approx 0.648 \quad \text{(Breakaway point between poles)}$$
$$\sigma_{in} \approx -2.082 \quad \text{(Break-in point to the left of zero)}$$

Step 5. Circular Shape of the Locus For a system with two poles and one zero, the complex portion of the root locus forms a circle.

– **Center:** The location of the zero, $c = -0.717$.
– **Radius:** The distance from the center to the breakaway point.

$$R = |0.648 - (-0.717)| = 1.365 \tag{9}$$

The equation of the circle in the complex z-plane is:

$$|z + 0.717| = 1.365 \tag{10}$$

Step 6. Stability Limit The system is stable when the roots lie inside the unit circle ($|z| < 1$). The locus crosses the unit circle when the gain is:

$$K_{crit} \approx 2.39 \tag{11}$$

For $K > 2.39$, the system becomes unstable.
The root locus is shown in the following figure.

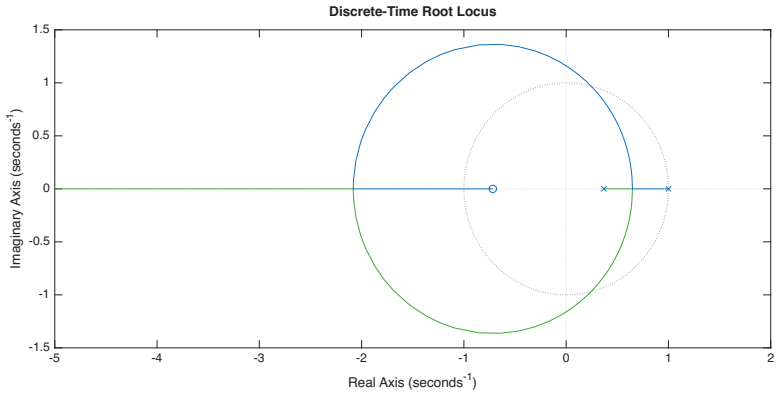


Figure 2: Root locus for question 1(D).

2(A). The continuous-time system is defined by

$$\begin{aligned} \dot{x}(t) &= Fx(t) + Gu(t), \\ y(t) &= Lx(t), \end{aligned} \tag{12}$$

with

$$F = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}, \quad G = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad L = \begin{bmatrix} 2 & 0 \end{bmatrix}.$$

Let the sampling time be T seconds.

(i) Discretize the continuous-time state-space equations (12). To this end, determine the matrices A, B, and C of the representation

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k), \\ y(k) &= Cx(k), \end{aligned} \tag{13}$$

when the input $u(t)$ is sampled and fed to a zero-order hold. [4 marks]
Hint: recall that $e^{Ft} = \mathcal{L}^{-1}[(sI-F)^{-1}]$, where $\mathcal{L}^{-1}$ indicates the inverse Laplace transform operator.
Solution: First, compute the state transition matrix $\Phi(t) = \mathcal{L}^{-1}\{(sI - F)^{-1}\}$:

$$\begin{aligned} (sI - F)^{-1} &= \begin{bmatrix} s & -1 \\ 0 & s+2 \end{bmatrix}^{-1} = \frac{1}{s(s+2)} \begin{bmatrix} s+2 & 1 \\ 0 & s \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{s} & \frac{1}{s(s+2)} \\ 0 & \frac{1}{s+2} \end{bmatrix} \end{aligned}$$

Using partial fraction expansion $\frac{1}{s(s+2)} = \frac{0.5}{s} - \frac{0.5}{s+2}$, we obtain:

$$\Phi(t) = \begin{bmatrix} 1 & 0.5(1 - e^{-2t}) \\ 0 & e^{-2t} \end{bmatrix}$$

Evaluating at $t = T$:

$$A = \Phi(T) = \begin{bmatrix} 1 & 0.5(1 - e^{-2T}) \\ 0 & e^{-2T} \end{bmatrix}$$

For B, we compute

$$B = \int_0^T \Phi(\tau) G \, d\tau = \begin{bmatrix} 0.5T - 0.25(1 - e^{-2T}) \\ 0.5(1 - e^{-2T}) \end{bmatrix}$$

.
Finally, $C = L = \begin{bmatrix} 2 & 0 \end{bmatrix}$.

(ii) Compute the input-output transfer function of the discrete-time system (13) from the matrices A, B, and C. [4 marks]
Solution: The transfer function is $G(z) = C(zI - A)^{-1}B$.

$$(zI - A)^{-1} = \frac{1}{(z-1)(z-\alpha)} \begin{bmatrix} z - \alpha & 0.5(1-\alpha) \\ 0 & z - 1 \end{bmatrix}, \quad \text{where } \alpha = e^{-2T}.$$

Pre-multiplying by $C = [2, 0]$ selects the first row scaled by 2:

$$C(zI - A)^{-1} = \frac{1}{(z-1)(z-\alpha)} \begin{bmatrix} 2(z-\alpha) & (1-\alpha) \end{bmatrix}$$

Multiplying by $B = [B_1 \; B_2]^\top$:

$$G(z) = \frac{2(z-\alpha)B_1 + (1-\alpha)B_2}{(z-1)(z-\alpha)}$$

Substituting B_1, B_2 leads to:

$$G(z) = \frac{2B_1}{z-1} + \frac{B_2(1-\alpha)}{(z-1)(z-\alpha)}$$

(iii) The continuous-time state-space system (12) has the input-output transfer function

$$\frac{Y(s)}{U(s)} = P(s) = L(sI - F)^{-1}G = \frac{2}{s(s+2)}.$$

Compute the equivalent discrete-time model $HP(z)$ for the plant $P(s)$ interconnected to the zero-order hold and the sampler. Show that $HP(z)$ is the same transfer function as the one obtained in part (ii). [4 marks]

Solution: We compute $HP(z) = (1 - z^{-1})\mathcal{Z}\{\mathcal{L}^{-1}[\frac{P(s)}{s}]\}$.

$$\frac{P(s)}{s} = \frac{2}{s^2(s+2)} = \frac{-0.5}{s} + \frac{1}{s^2} + \frac{0.5}{s+2}$$

Taking the Z-transform:

$$\mathcal{Z}\left\{\frac{P(s)}{s}\right\} = \frac{-0.5z}{z-1} + \frac{Tz}{(z-1)^2} + \frac{0.5z}{z - e^{-2T}}$$

Multiplying by $\frac{z-1}{z}$:

$$HP(z) = -0.5 + \frac{T}{z-1} + \frac{0.5(z-1)}{z - e^{-2T}}$$

Rearranging terms confirms this is identical to the result in part (ii).

(iv) Calculate the controllability matrix $\mathcal{C}$ for the continuous-time system (F, G) and determine its rank. Based on this result, explain whether the equivalent discrete-time system (A, B) is controllable for all sampling times $T > 0$. [4 marks]

Solution: The continuous controllability matrix is $\mathcal{C}_{cont} = [G \quad FG]$.

$$FG = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \implies \mathcal{C}_{cont} = \begin{bmatrix} 0 & 1 \\ 1 & -2 \end{bmatrix}$$

The rank is 2, so the continuous system is controllable.

Sampling preserves controllability unless the sampling frequency is "pathological" ($\lambda_i - \lambda_j = \frac{2\pi kj}{T}$). Since the eigenvalues of F (0 and -2) are distinct real numbers, their difference is real. Therefore, the discrete system is controllable for all $T > 0$.

(v) Calculate the observability matrix $\mathcal{O}$ for the continuous-time system using the output matrix $L = \begin{bmatrix} 2 & 0 \end{bmatrix}$. Determine if the system is observable. Does the discrete-time system (A, C) retain this property? [4 marks]

Solution: The continuous observability matrix is $\mathcal{O}_{cont} = [L^T \quad F^T L^T]^T$.

$$LF = \begin{bmatrix} 2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 2 \end{bmatrix} \implies \mathcal{O}_{cont} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

The rank is 2, so the system is observable. This property is preserved in the discrete domain for all $T > 0$.

2(B). Consider the discrete-time system (13) with

$$A = \begin{bmatrix} 1 & 0.4 \\ 0 & 0.6 \end{bmatrix}, \quad B = \begin{bmatrix} 0.1 \\ 0.4 \end{bmatrix}$$

Design a state feedback control law $u[k] = -Kx[k]$ with gain matrix $K = \begin{bmatrix} k_1 & k_2 \end{bmatrix}$ to place the closed-loop poles at the complex locations $z_{1,2} = 0.5 \pm j0.2$. [5 marks]

Solution: Desired characteristic equation:

$$(z - 0.5 - j0.2)(z - 0.5 + j0.2) = (z - 0.5)^2 + 0.04 = z^2 - z + 0.29$$

The closed-loop matrix is $A_{cl} = A - BK$:

$$A_{cl} = \begin{bmatrix} 1 & 0.4 \\ 0 & 0.6 \end{bmatrix} - \begin{bmatrix} 0.1k_1 & 0.1k_2 \\ 0.4k_1 & 0.4k_2 \end{bmatrix} = \begin{bmatrix} 1 - 0.1k_1 & 0.4 - 0.1k_2 \\ -0.4k_1 & 0.6 - 0.4k_2 \end{bmatrix}$$

We equate the Trace and Determinant to the desired polynomial coefficients:

1. $\text{Trace}(A_{cl}) = 1.6 - 0.1k_1 - 0.4k_2 = 1 \implies 0.1k_1 + 0.4k_2 = 0.6$
2. $\det(A_{cl}) = 0.6 + 0.1k_1 - 0.4k_2 = 0.29 \implies 0.1k_1 - 0.4k_2 = -0.31$

Adding the two equations: $0.2k_1 = 0.29 \implies k_1 = 1.45$.
Substituting back: $0.145 + 0.4k_2 = 0.6 \implies 0.4k_2 = 0.455 \implies k_2 = 1.1375$.

$$K = \begin{bmatrix} 1.45 & 1.1375 \end{bmatrix}$$

3(A). Consider a scalar linear discrete-time system given by the dynamics:

$$x(k+1) = ax(k) + bu(k) + w(k)$$

where $x(k) \in \mathbb{R}$ is the state, $u(k) \in \mathbb{R}$ is the control input, and $w(k)$ is an independent and identically distributed (i.i.d.) Gaussian noise sequence with $w(k) \sim \mathcal{N}(0, \sigma^2)$. Here, $a, b \neq 0$.

Give a finite horizon N, the objective is to minimize the finite-horizon cost function:

$$J = \mathbb{E}\left[sx^2(N) + \sum_{k=0}^{N-1}(qx^2(k) + ru^2(k))\right]$$

where $s, q \geq 0$ and $r > 0$.

(i) Define the cost-to-go function (value function) $V_k(x(k))$ for the time step k. [2 marks]

Solution. The cost-to-go function (or value function) $V_k(x(k))$ represents the minimum expected cost accumulated from time step k to the terminal time N, given that the system is in state $x(k)$ at time k. It is defined as:

$$V_k(x(k)) = \min_{u_k,\ldots,u_{N-1}} \mathbb{E}\left[sx^2(N) + \sum_{i=k}^{N-1}(qx^2(i) + ru^2(i)) \,\middle|\, x(k)\right] \tag{14}$$

At the terminal time $k = N$, the cost is determined solely by the boundary condition:

$$V_N(x(N)) = sx^2(N) \tag{15}$$

(ii) Derive, using the dynamic programming, the optimal control $u(k)$ that minimizes $J(x(0))$ and determine the closed-form expression for the minimal total cost $J_{\min}(x(0))$ (i.e., $V_0(x(0))$). [6 marks]

Solution. We apply Dynamic Programming (DP) working backward from $k = N$ to $k = 0$.

Step 1: Using the Principle of Optimality, the value function satisfies the recursive Bellman equation:

$$V_k(x(k)) = \min_{u(k)} \left\{qx^2(k) + ru^2(k) + \mathbb{E}\left[V_{k+1}(x(k+1)) \mid x(k)\right]\right\} \tag{16}$$

Step 2: We assume the value function is quadratic of the form:

$$V_k(x(k)) = P_k x^2(k) + \alpha_k \tag{17}$$

where P_k is a deterministic scalar (the Riccati variable) and α_k is a scalar constant accounting for the stochastic noise cost. Comparing with $V_N(x(N)) = sx^2(N)$, we get:

$$P_N = s, \quad \alpha_N = 0$$

Step 3: Substitute the system dynamics $x(k+1) = ax(k) + bu(k) + w(k)$ into the assumed form for V_{k+1}:

$$\begin{aligned}\mathbb{E}[V_{k+1}(x(k+1))] &= \mathbb{E}\left[P_{k+1}(ax(k) + bu(k) + w(k))^2 + \alpha_{k+1}\right] \\ &= \mathbb{E}\left[P_{k+1}((ax(k) + bu(k))^2 + 2(ax(k) + bu(k))w(k) + w^2(k)) + \alpha_{k+1}\right]\end{aligned}$$

Since $w(k) \sim \mathcal{N}(0, \sigma^2)$, we have $\mathbb{E}[w(k)] = 0$ and $\mathbb{E}[w(k)^2] = \sigma^2$. The expectation becomes:

$$\mathbb{E}[V_{k+1}] = P_{k+1}(ax(k) + bu(k))^2 + P_{k+1}\sigma^2 + \alpha_{k+1} \tag{18}$$

Step 4: Substitute the expectation back into the Bellman equation (16):

$$V_k(x) = \min_{u_k} \left\{qx^2(k) + ru^2(k) + P_{k+1}(ax(k) + bu(k))^2 + P_{k+1}\sigma^2 + \alpha_{k+1}\right\} \tag{19}$$

To minimize, differentiate the expression inside the braces with respect to u_k and set to zero:

$$\frac{\partial}{\partial u(k)}(\dots) = 2ru(k) + 2P_{k+1}b(ax(k) + bu(k)) = 0$$
$$0 = u(k)(r + P_{k+1}b^2) + P_{k+1}abx(k)$$

Solving for the optimal control $u^*(k)$:

$$\boxed{u^*(k) = -\frac{P_{k+1}ab}{r + P_{k+1}b^2}x(k)} \tag{20}$$

Step 5: Substitute $u^*(k)$ back into the expression for $V_k(x)$ to find the recurrence for P_k and α_k. After algebraic simplification, the quadratic coefficient P_k follows the discrete-time algebraic Riccati equation:

$$P_k = q + P_{k+1}a^2 - \frac{(P_{k+1}ab)^2}{r + P_{k+1}b^2} \tag{21}$$

The constant term recurrence is:

$$\alpha_k = \alpha_{k+1} + P_{k+1}\sigma^2 \tag{22}$$

Iterating α_k backward from $\alpha_N = 0$, we get $\alpha_0 = \sum_{k=0}^{N-1} P_{k+1}\sigma^2$.
The minimal total cost is $V_0(x(0))$:

$$\boxed{J_{\min}(x(0)) = P_0x^2(0) + \sigma^2 \sum_{k=1}^{N} P_k} \tag{23}$$

4(B). Consider the discrete–time LTI system

$$x(k+1) = Ax(k) + Bu(k), \qquad y(k) = Cx(k) + Du(k), \tag{24}$$

where $x(k) \in \mathbb{R}^n$, $u(k) \in \mathbb{R}^m$, and $y(k) \in \mathbb{R}^p$. Assume that the pair (A, B) is controllable.
Let $\{u_d(k), y_d(k)\}_{k=0}^{T-1}$ be a measured input–output trajectory generated by the system (24). For a signal $z(k) \in \mathbb{R}^q$, define the block Hankel matrix

$$Z_{0,t,T-t+1} = \begin{bmatrix} z(0) & z(1) & \cdots & z(T-t) \\ z(1) & z(2) & \cdots & z(T-t+1) \\ \vdots & \vdots & \ddots & \vdots \\ z(t-1) & z(t) & \cdots & z(T-1) \end{bmatrix}.$$

(i) Define what it means for the input $u_d[0, T-1]$ to be *persistently exciting of order t.* [3 marks]
Solution. The input $u_d[0, T-1]$ is persistently exciting of order t if

$$\operatorname{rank}(U_{0,t,T-t+1}) = mt.$$

This means that the rows of the Hankel matrix span all possible length–t input sequences in $\mathbb{R}^{mt}$.

(ii) Assume that $u_d[0, T-1]$ is persistently exciting of order $n + t$. Show that for every length–t input–output trajectory $\{u(k), y(k)\}_{k=0}^{t-1}$ of system (24), there exists a vector $g \in \mathbb{R}^{T-t+1}$ such that

$$\begin{bmatrix} U_{0,t,T-t+1} \\ Y_{0,t,T-t+1} \end{bmatrix} g = \begin{bmatrix} u(0) \\ \vdots \\ u(t-1) \\ y(0) \\ \vdots \\ y(t-1) \end{bmatrix}. \tag{25}$$

Hint: If (A, B) is controllable and the input $u_{d,[0,T-1]}$ is persistently exciting of order $n+t$, then the stacked matrix has full row rank:

$$rank \begin{bmatrix} U_{0,t,T-t+1} \\ X_{0,T-t+1} \end{bmatrix} = n + tm.$$

[5 marks]

Solution. The system input–output response of over $[0, t-1]$ can be expressed as

$$\begin{bmatrix} u_{[0,t-1]} \\ y_{[0,t-1]} \end{bmatrix} = \begin{bmatrix} I_{tm} & \mathbb{O}_{tm\times n} \\ \mathcal{T}_t & \mathcal{O}_t \end{bmatrix} \begin{bmatrix} u_{[0,t-1]} \\ x_0 \end{bmatrix}$$

where

$$\mathcal{T}_t := \begin{bmatrix} D & 0 & \cdots & 0 \\ CB & D & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ CA^{t-2}B & CA^{t-3}B & \cdots & D \end{bmatrix}, \quad \mathcal{O}_t := \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{t-1} \end{bmatrix}$$

Let

$$\begin{bmatrix} U_{0,t,T-t+1} \\ Y_{0,t,T-t+1} \end{bmatrix} := \left[\begin{array}{cccc} u_d(0) & u_d(1) & \cdots & u_d(T-t) \\ \vdots & \vdots & \ddots & \vdots \\ u_d(t-1) & u_d(t) & \cdots & u_d(T-1) \\ \hline y_d(0) & y_d(1) & \cdots & y_d(T-t) \\ \vdots & \vdots & \ddots & \vdots \\ y_d(t-1) & y_d(t) & \cdots & y_d(T-1) \end{array}\right]$$

the corresponding Hankel matrix. One can rewrite

$$\begin{bmatrix} U_{0,t,T-t+1} \\ Y_{0,t,T-t+1} \end{bmatrix} = \begin{bmatrix} I_{tm} & \mathbb{O}_{tm\times n} \\ \mathcal{T}_t & \mathcal{O}_t \end{bmatrix} \begin{bmatrix} U_{0,t,T-t+1} \\ X_{0,T-t+1} \end{bmatrix}$$

where the state sequence matrix is defined as:

$$X_{0,T-t+1} := \begin{bmatrix} x_d(0) & x_d(1) & \cdots & x_d(T-t) \end{bmatrix}$$

If the input $u_{d,[0,T-1]}$ is persistently exciting of order $n+t$, then the stacked matrix has full row rank:

$$\text{rank} \begin{bmatrix} U_{0,t,T-t+1} \\ X_{0,T-t+1} \end{bmatrix} = n + tm.$$

This rank condition implies the existence of a vector $g \in \mathbb{R}^{T-t+1}$ such that:

$$\begin{bmatrix} u_{[0,t-1]} \\ x_0 \end{bmatrix} = \begin{bmatrix} U_{0,t,T-t+1} \\ X_{0,T-t+1} \end{bmatrix} g.$$

By replacing this expression into the system response equation, we get:

$$\begin{aligned} \begin{bmatrix} u_{[0,t-1]} \\ y_{[0,t-1]} \end{bmatrix} &= \begin{bmatrix} I_{tm} & \mathbb{O}_{tm\times n} \\ \mathcal{T}_t & \mathcal{O}_t \end{bmatrix} \begin{bmatrix} U_{0,t,T-t+1} \\ X_{0,T-t+1} \end{bmatrix} g \\ &= \begin{bmatrix} U_{0,t,T-t+1} \\ Y_{0,t,T-t+1} \end{bmatrix} g. \end{aligned}$$

(iii) Let $X_{0,T} := [\, x_d(0) \ \cdots \ x_d(T-1) \,]$ and $X_{1,T} := [\, x_d(1) \ \cdots \ x_d(T) \,]$. Assume that

$$\text{rank} \begin{bmatrix} U_{0,1,T} \\ X_{0,T} \end{bmatrix} = m + n.$$

Show that a data-driven representation of the open-loop system dynamics is

$$x(k+1) = X_{1,T} \begin{bmatrix} U_{0,1,T} \\ X_{0,T} \end{bmatrix}^{\dagger} \begin{bmatrix} u(k) \\ x(k) \end{bmatrix}$$

where $[\cdot]^{\dagger}$ is the Moore-Penrose inverse. [5 marks]

Solution. Recall that $X_{1,T} = \begin{bmatrix} B & A \end{bmatrix} \begin{bmatrix} U_{0,1,T} \\ X_{0,T} \end{bmatrix}$.

Thus,

$$\begin{bmatrix} B & A \end{bmatrix} = X_{1,T} \begin{bmatrix} U_{0,1,T} \\ X_{0,T} \end{bmatrix}^{\dagger}$$

Then, we have

$$x(k+1) = \begin{bmatrix} B & A \end{bmatrix} \begin{bmatrix} u(k) \\ x(k) \end{bmatrix} = X_{1,T} \begin{bmatrix} U_{0,1,T} \\ X_{0,T} \end{bmatrix}^{\dagger} \begin{bmatrix} u(k) \\ x(k) \end{bmatrix}.$$